

Gurgaon Real Estate Data Analysis

Turning Raw Property Data into Business Insights | Exploratory Data Analysis (EDA)



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 Python

 Pandas

 Matplotlib

 Seaborn

Project Overview

Objective

Analyze residential property listings across Gurgaon to uncover key market trends and answer real business questions for buyers, investors, and developers.

Dataset Features

 Property Price	 Area (Sqft)
 Locality	 Property Type
 BHK Configuration	 Builder
 RERA Approval	 Rate per Sqft



Data Collection



Data Cleaning



Exploratory EDA



Business Insights

Data Cleaning & Preparation

- ✓ Standardized column names to lower_snake_case
- ✓ Removed duplicate records across raw property rows
- ✓ Converted price, area, & rate/sqft into numeric format
- ✓ Cleaned & trimmed categorical columns (status, rera, type)
- ✓ Prepared clean dataframe ready for Pandas aggregations

Executed in Python 3 environment



main.py - Cleaning Pipeline

```
# Standardize column headers
df.columns = df.columns.str.strip().str.lower()
df = df.drop_duplicates()

# Numeric columns cleaning
df['price'] = df['price'].astype(str).str.replace(',', '').astype(float)
df['area'] = df['area'].astype(str).str.replace(',', '').astype(int)
df['rate_per_sqft'] = df['rate_per_sqft'].astype(str).replace(',', '',
').astype(int)
```

 Python 3.x

 Pandas Wrangling

? Business Questions Answered



Which is the costliest property in the dataset?



Which locality has the highest average property price?



Which locality commands the highest rate per square foot?



Do ready-to-move properties cost more than under construction?



Do RERA-approved properties command a price premium?



How strongly does area (sqft) impact overall property price?



Which BHK configuration is the most expensive on average?



Which property type (Apartment, Villa, Independent) is highest?



Which developers & builders consistently price higher?



Are larger homes always more expensive per square foot?

Analysis Results & Output

 Costliest Property

₹12.26 Cr

Apartment located in **Sector 42**

 Highest Avg Price

Baliawas

Locality with top overall average listing price

 Highest Rate/Sqft

Sector 42

Prime locality commanding maximum rate per sqft

>_ Terminal Output Execution – main.py

Python 3.11

```
utkarshkarhana@MacBook-Air % python3 main.py
```

```
The costliest flat in the dataset is: apartment located in Sector 42 with a price of 122,630,000.0
```

```
The locality with the highest average price is: Baliawas
```

```
The locality with the highest rate per square foot is: Sector 42
```


Additional Results Breakdown

Ready-to-Move vs Under Construction

Ready-to-Move > Under Construction

Ready-to-move properties command a clear price premium in the current Gurgaon market.

RERA vs Non-RERA Approval

No Premium for RERA

RERA-approved properties do not automatically command higher price per sqft.

Most Expensive Property Type

Villa

Villas are the costliest property type on a rate per square foot basis.

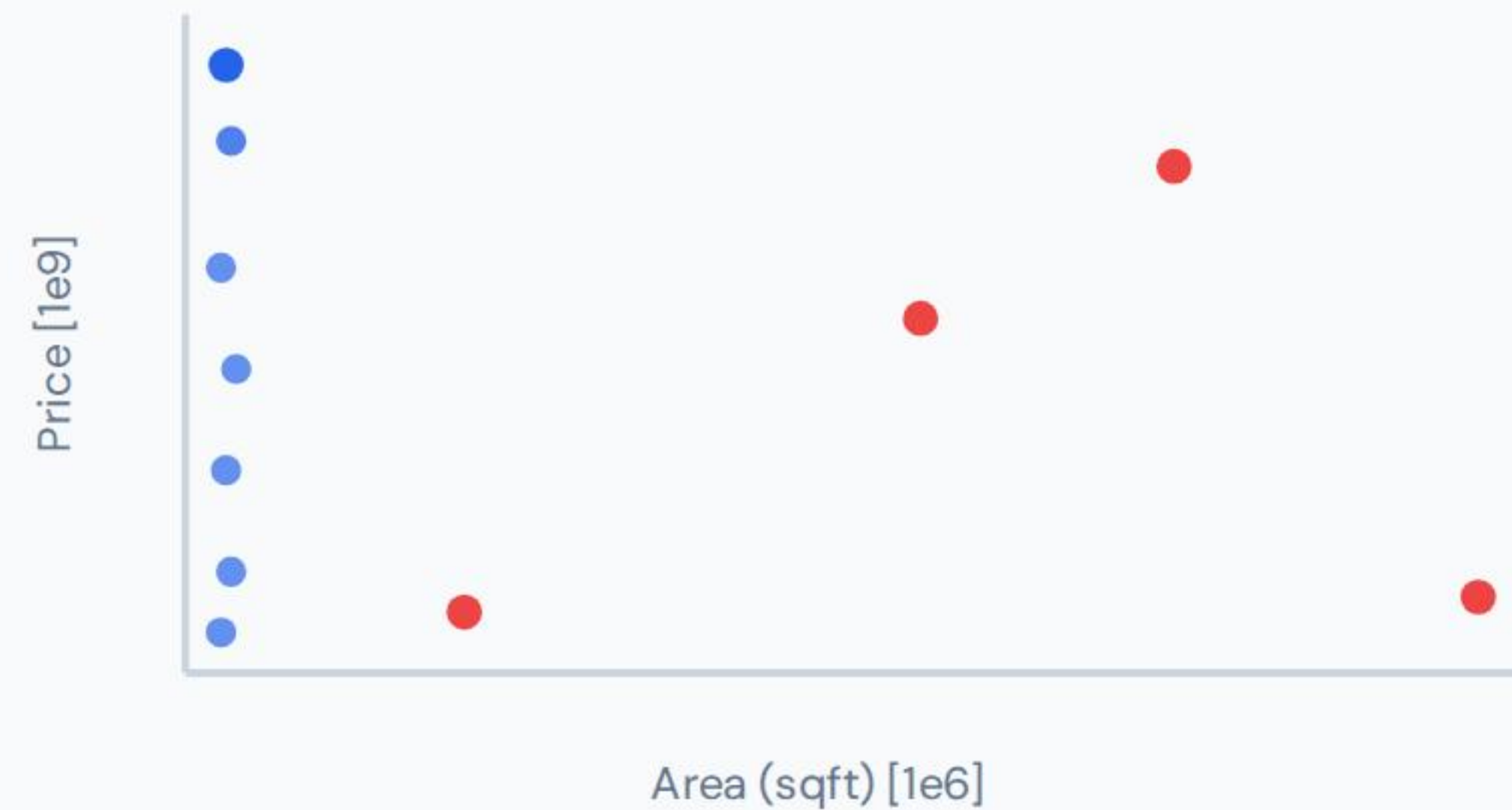
Top 5 Premium Builders

Camelliaass, Cameliaas, Tulip, Prom, Magnoliaass

Builders identified as charging the highest average rate per sqft across listings.

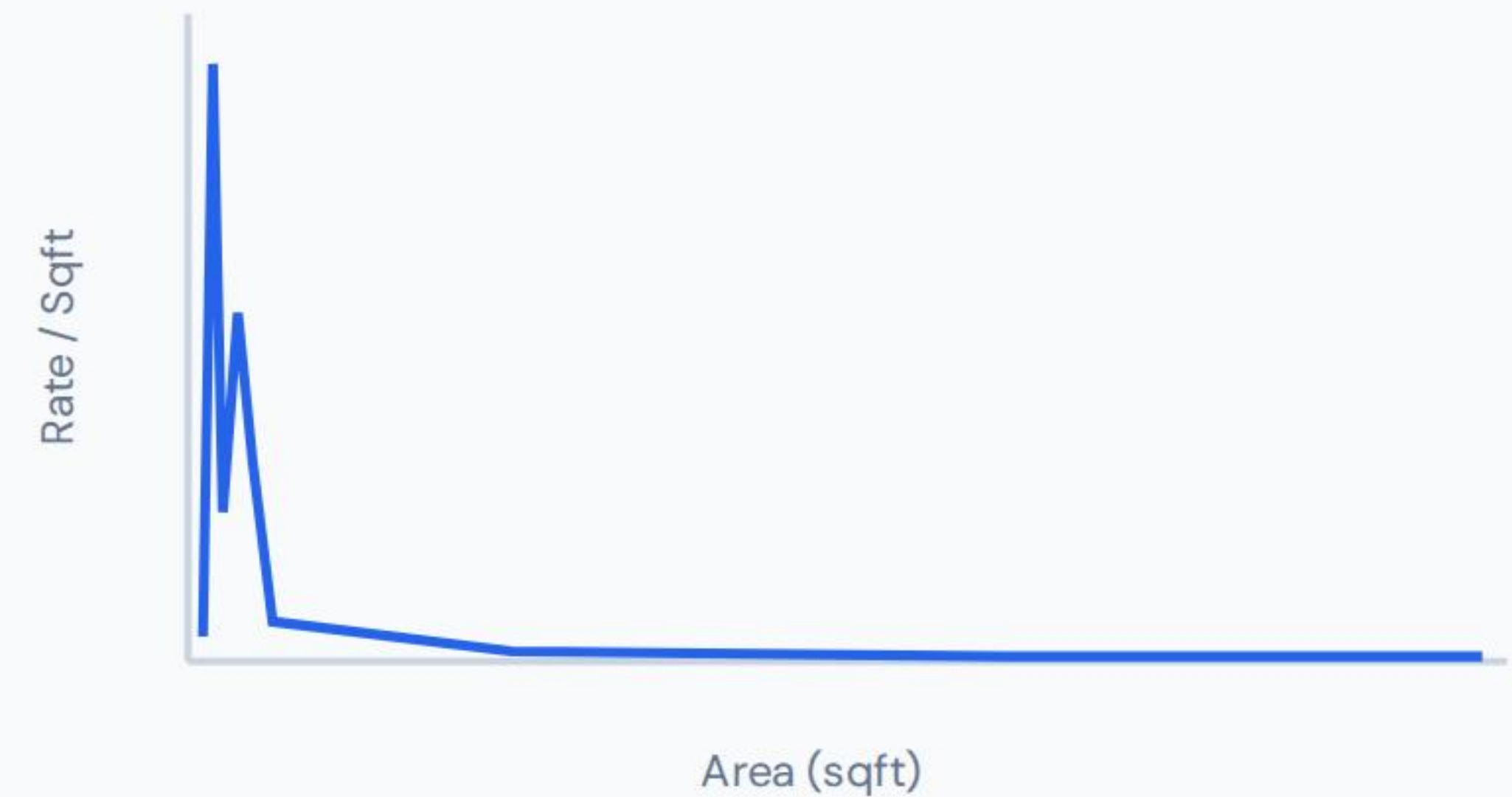
Visual Analysis & Matplotlib Plots

Area vs Price Scatter Plot



Observation: Prices generally scale with area, with a strong vertical concentration in core sizes and clear luxury outliers.

Area vs Rate per Sqft



Observation: Larger homes are not always more expensive per sqft. High rate/sqft spikes occur mainly in exclusive micro-markets.

Key Business Insights



Baliavas emerges as the locality with the highest average property price across Gurgaon listings.



Sector 42 commands the highest rate per sqft & holds the costliest apartment (₹12.26 Cr).



Ready-to-move properties cost significantly more upfront compared to under-construction projects.



Villas & Independent Houses command the highest average rate per square foot overall.



Premium Builders (Camelliaass, Tulip, Magnoliaass) charge a notable brand markup over market averages.



Data Anomaly: Dataset contains an unusual 114 BHK entry—a clear outlier requiring domain validation before business decisions.

Skills Applied

 Python	 Pandas	 Data Cleaning	 Data Wrangling
 EDA	 Matplotlib	 Seaborn	 Business Analysis
 Data Visualization	 Problem Solving	 GroupBy & Aggs	 Critical Thinking



Project Takeaways

This project strengthened my understanding of real-world data analysis by transforming raw property listings into actionable business intelligence.

Key Learnings:

✓ Data Cleaning

✓ Exploratory Data Analysis

✓ Data Storytelling

✓ Business-Oriented Thinking

✓ Python for Data Analysis

"Every dataset tells a story. My goal is to uncover it through data."

Utkarsh Karhana

Data Analyst | Open to opportunities

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